

Lecture 20

Bayesian Linear Models Part 2

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(Edited from the lecture notes by Brian J. Reich, NC State)



Recap: Linear regression

- ▶ Linear regression is by far the most common statistical model.
- ▶ It includes as special cases the t-test and ANOVA.
- ▶ The multiple linear regression model is

$$Y_i \sim \text{Normal}(\beta_0 + X_{i1}\beta_1 + \dots + X_{ip}\beta_p, \sigma^2),$$

independently across the $i = 1, \dots, n$ observations.

- ▶ As we'll see, Bayesian and classical linear regression are similar if $n \gg p$ and the priors are non-informative.
- ▶ However, the results can be different for challenging problems, and the interpretation is different in all cases.

Recap: Review of least squares

- ▶ The least squares estimate of $\beta = (\beta_0, \beta_1, \dots, \beta_p)^T$ is

$$\hat{\beta}_{OLS} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^n (Y_i - \mu_i)^2,$$

where $\mu_i = \beta_0 + X_{i1}\beta_1 + \dots + X_{ip}\beta_p$.

- ▶ $\hat{\beta}_{OLS}$ is unbiased even if the errors are non-Gaussian.
- ▶ If the errors are Gaussian, then the likelihood is proportional to

$$\prod_{i=1}^n \exp \left[-\frac{(Y_i - \mu_i)^2}{2\sigma^2} \right] = \exp \left[-\frac{\sum_{i=1}^n (Y_i - \mu_i)^2}{2\sigma^2} \right].$$

- ▶ Therefore, if the errors are Gaussian $\hat{\beta}_{OLS}$ is also the MLE.

Recap: Review of least squares

- ▶ Linear regression is often simpler to describe using linear algebra notation.
- ▶ Let $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ be the response vector and $\mathbf{X}$ be the $n \times (p + 1)$ matrix of covariates.
- ▶ Then the mean of $\mathbf{Y}$ is $\mathbf{X}\beta$ and the least squares solution is

$$\hat{\beta}_{OLS} = \underset{\beta}{\operatorname{argmin}} (\mathbf{Y} - \mathbf{X}\beta)^T (\mathbf{Y} - \mathbf{X}\beta) = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}.$$

- ▶ If the errors are Gaussian then the sampling distribution is

$$\hat{\beta}_{OLS} \sim \text{Normal} \left[\beta, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \right].$$

- ▶ If the variance σ^2 is estimated using the mean squared residual error, then the sampling distribution is multivariate t .

Recap: Bayesian regression

- ▶ The likelihood remains

$$Y_i \sim \text{Normal}(\beta_0 + X_{i1}\beta_1 + \dots + X_{ip}\beta_p, \sigma^2)$$

independent for $i = 1, \dots, n$ observations.

- ▶ As with a least squares analysis, it is crucial to verify this is appropriate using qq-plots, added variable plots, etc.
- ▶ A Bayesian analysis also requires priors for β and σ .
- ▶ We will focus on prior specification since this piece is uniquely Bayesian.

Priors

- ▶ For the purpose of setting priors, it is helpful to standardize both the response and each covariate to have mean zero and variance one.
- ▶ Many priors for β have been considered in the literature:
 1. Improper priors
 2. Gaussian priors
 3. Double exponential priors
 4. Many, many more...

Improper priors

- ▶ With σ fixed, the Jeffreys' prior is flat $p(\beta) = 1$.
- ▶ This is improper, but the posterior is proper under the same conditions required by least squares.
- ▶ If σ is known then

$$\beta|\mathbf{Y} \sim \text{Normal} \left[\hat{\beta}_{OLS}, \sigma^2(\mathbf{X}^T \mathbf{X})^{-1} \right].$$

- ▶ Therefore, the results should be similar to least squares.
- ▶ How are they different?

Improper priors

- ▶ Of course we rarely know σ .
- ▶ A conjugate uninformative prior is

$$\sigma^2 \sim \text{InvGamma}(a, b),$$

with a and b set to be small, say $a = b = 0.01$.

- ▶ In this case, the posterior of β follows a multivariate t centered on $\hat{\beta}_{OLS}$.
- ▶ Again, the results are similar to OLS.

Improper priors

- ▶ The objective Bayes Jeffreys prior is

$$p(\beta, \sigma^2) = \left(\frac{1}{\sigma^2} \right)^{p/2+1},$$

which is the inverse gamma prior with $a = p/2$ and $b \rightarrow 0$.

- ▶ This gives posterior (marginal over σ^2)

$$\beta | \mathbf{Y} \sim t_n \left(\hat{\beta}_{OLS}, \hat{\sigma}^2 (\mathbf{X}^T \mathbf{X})^{-1} \right),$$

where $\hat{\sigma}^2 = (\mathbf{Y} - \mathbf{X}\hat{\beta}_{OLS})^T (\mathbf{Y} - \mathbf{X}\hat{\beta}_{OLS}) / n$.

- ▶ The posterior is proper in the same situations that the least squares solution exists.

Multivariate normal prior

- ▶ Another common prior for is Zellner's g-prior

$$\beta \sim \text{Normal} \left[0, \frac{\sigma^2}{g} (\mathbf{X}^T \mathbf{X})^{-1} \right].$$

- ▶ This prior is proper assuming $\mathbf{X}$ is full rank.

- ▶ The posterior mean is

$$\frac{1}{1+g} \hat{\beta}_{OLS}.$$

- ▶ This shrinks the least estimate towards zero.
- ▶ g controls the amount of shrinkage.
- ▶ $g = 1/n$ is common, and called the unit information prior.

Univariate Gaussian priors

- ▶ If there are many covariates or the covariates are collinear, then $\hat{\beta}_{OLS}$ is unstable.
- ▶ Independent priors can counteract collinearity

$$\beta_j \sim \text{Normal}(0, \sigma^2/g)$$

independent over j .

- ▶ The posterior mode is

$$\underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^n (Y_i - \mu_i)^2 + g \sum_{j=1}^p \beta_j^2 \right\}.$$

- ▶ In classical statistics, this is known as the ridge regression solution and is used to stabilize the least squares solution.

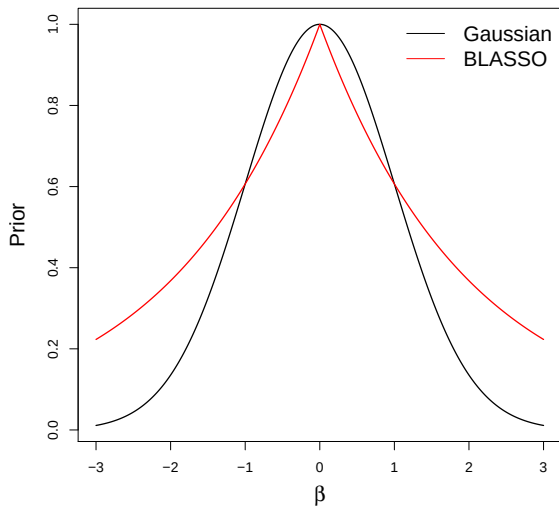
BLASSO

- ▶ An increasingly-popular prior is the double exponential or Bayesian LASSO prior.
- ▶ The prior is $\beta_j \sim \text{DE}(\tau)$ which has PDF

$$f(\beta) \propto \exp\left(-\frac{|\beta|}{\tau}\right).$$

- ▶ The square in the Gaussian prior is replaced with an absolute value.
- ▶ The shape of the PDF is thus more peaked at zero (next slide).
- ▶ The BLASSO prior favors settings where there are many β_j near zero and a few large β_j .
- ▶ That is, p is large but most of the covariates are noise.

BLASSO



BLASSO

- ▶ The posterior mode is

$$\underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^n (Y_i - \mu_i)^2 + g \sum_{j=1}^p |\beta_j| \right\}$$

- ▶ In classical statistics, this is known as the LASSO solution.
- ▶ It is popular because it adds stability by shrinking estimates towards zero, and also sets some coefficients to zero.
- ▶ Covariates with coefficients set to zero can be removed.
- ▶ Therefore, LASSO performs variables selection and estimation simultaneously.

Computing

- ▶ With flat or Gaussian (with fixed prior variance) priors the posterior is available in closed-form and Monte Carlo sampling is not needed.
- ▶ JAGS also works well, but there are $\mathbb{R}$ packages dedicated just to Bayesian linear regression that are preferred for big/hard problems.
- ▶ BLR is probably the most common.

Computing for the BLASSO

- ▶ For the BLASSO prior, the full conditionals are more complicated.
- ▶ There is a trick to make all full conditionals conjugate so that Gibbs sampling can be used.
- ▶ Metropolis sampling works fine too.
- ▶ BLR works well for BLASSO and is super fast.

Summarizing the results

- ▶ The standard summary is a table with marginal means and 95% intervals for each β_j .
- ▶ This becomes unwieldy for large p .
- ▶ Picking a subset of covariates is a crucial step in a linear regression analysis.
- ▶ We will discuss this later in the course.
- ▶ Common methods include cross-validation, information criteria, and stochastic search.

Predictions

- ▶ Say we have a new covariate vector $\mathbf{X}_{new}$ and we would like to predict the corresponding response Y_{new} .

- ▶ A plug-in approach would fix β and σ at their posterior means $\hat{\beta}$ and $\hat{\sigma}$ to make predictions

$$Y_{new} | \hat{\beta}, \hat{\sigma} \sim \text{Normal}(\mathbf{X}_{new} \hat{\beta}, \hat{\sigma}^2).$$

- ▶ However this plug-in approach suppresses uncertainty about β and σ .
- ▶ Therefore these prediction intervals will be slightly too narrow leading to undercoverage.

Posterior predictive distribution (PPD)

- ▶ We should really account for all uncertainty when making predictions, including our uncertainty about β and σ .
- ▶ We really want the PPD

$$\begin{aligned} p(Y_{new}|\mathbf{Y}) &= \int f(Y_{new}, \beta, \sigma | \mathbf{Y}) d\beta d\sigma \\ &= \int f(Y_{new} | \beta, \sigma) f(\beta, \sigma | \mathbf{Y}) d\beta d\sigma. \end{aligned}$$

- ▶ Marginalizing over the model parameters accounts for their uncertainty.
- ▶ The concept of the PPD applies generally (e.g., logistic regression) and means the distribution of the predicted value marginally over model parameters.

Posterior predictive distribution (PPD)

- ▶ MCMC naturally gives draws from Y_{new} 's PPD

- ▶ For MCMC iteration t we have $\beta^{(t)}$ and $\sigma^{(t)}$

- ▶ For MCMC iteration t we sample

$$Y_{new}^{(t)} \sim \text{Normal}(\mathbf{X}\beta^{(t)}, \sigma^{(t)2})$$

- ▶ $Y_{new}^{(1)}, \dots, Y_{new}^{(S)}$ are samples from the PPD

- ▶ This is an example of the claim that “Bayesian methods naturally quantify uncertainty”.

Thank you!